

SpaceX Falcon 9 First Stage Landing Prediction

Data Science Capstone Project Report

IBM Applied Data Science Capstone · Coursera

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GitHub: <https://github.com/pankh-ui/IBM-Applied-Data-Science-Capstone>

Project Repository

All notebooks, the dashboard script and this report

<https://github.com/pankh-ui/IBM-Applied-Data-Science-Capstone>

- 1-jupyter-labs-spacex-data-collection-api.ipynb — SpaceX REST API data collection
- 2-jupyter-labs-webscraping.ipynb — BeautifulSoup scraping of the Wikipedia launch table
- 3-labs-jupyter-spacex-Data-wrangling.ipynb — landing outcome converted to a binary label
- 4-jupyter-labs-eda-dataviz.ipynb — exploratory charts and one-hot feature engineering
- 5-jupyter-labs-eda-sql-coursera_sqlite.ipynb — the ten SQL exploration queries
- 6-lab-jupyter-launch-site-location.ipynb — Folium site, outcome and proximity maps
- 7-SpaceX-Machine-Learning-Prediction.ipynb — four tuned classifiers and their evaluation
- spacex_dash_app.py — the Plotly Dash launch records dashboard
- Data_Science_Capstone_Project_Report.pdf — this presentation

Outline

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Executive Summary

Methods used and the key results, in brief

- Goal: predict whether the Falcon 9 first stage will land successfully, so that the cost of a launch can be estimated. A reused first stage takes a launch from roughly \$165M to about \$62M.
- Methodologies used:
 - Data collection from the SpaceX REST API (JSON → pandas) and web scraping of the Falcon 9 launch table on Wikipedia with BeautifulSoup
 - Data wrangling: null handling, filtering to Falcon 9, and converting eight landing outcomes into a binary Class label (1 = landed, 0 = did not land)
 - EDA with pandas / Matplotlib / Seaborn and with SQL (SQLite magic commands)
 - Interactive visual analytics with Folium maps and a Plotly Dash dashboard
 - Predictive analysis: Logistic Regression, SVM, Decision Tree and KNN, each tuned with GridSearchCV (10-fold CV) on standardized features
- Key results:
 - Launch success has risen steadily from 0% in 2010 to roughly 80% by 2019-2020
 - KSC LC-39A has the highest success rate of any site; ES-L1, GEO, HEO and SSO orbits show 100% success in this dataset
 - Low-weight payloads (< 4000 kg) land successfully more often than heavy ones
 - All four models reach the same test accuracy of 83.33%; the Decision Tree has the best validation score (86.25%), and the confusion matrix shows false positives are the only meaningful error mode

Introduction

Project background and the problem being solved

- **Background:**

- SpaceX advertises Falcon 9 launches at \$62 million, while other providers charge upward of \$165 million. The saving comes from re-using the first stage.
- If the first stage does not land, the booster is lost and that launch costs far more. Predicting the landing therefore predicts the cost of the launch.
- Space Y, a competitor, wants to bid against SpaceX and needs that cost model.

- **Problems we want to answer:**

- What factors determine whether the first stage lands successfully?
- How do payload mass, launch site, orbit type and booster version interact with the landing outcome?
- Has the landing success rate improved over time?
- Which classification model best predicts a successful landing, and how accurate is it?

Methodology

How the data was collected, cleaned, explored and modeled

- Data collection methodology — SpaceX REST API requests and Wikipedia web scraping
- Data wrangling methodology — null handling and binary landing-outcome labeling
- EDA and data visualization methodology — the charts plotted and why each was chosen
- EDA with SQL methodology — the ten queries run against the SQLite database
- Interactive visual analytics methodology — Folium map objects and Dash components
- Predictive analysis (classification) methodology — standardizing, splitting, tuning and evaluating four models

Methodology — Overall Workflow

- 1. Data collection — SpaceX REST API (api.spacexdata.com/v4/launches/past) and web scraping of the Wikipedia Falcon 9 launch records table.
- 2. Data wrangling — filter to Falcon 9, impute missing PayloadMass with the column mean, and convert the eight landing outcomes into a binary Class label.
- 3. Exploratory data analysis — Seaborn / Matplotlib charts and SQL queries (SQLite via %sql magic) to find the relationships that matter.
- 4. Interactive visual analytics — Folium maps of every launch site and their proximities, plus a Plotly Dash dashboard with a site dropdown and a payload slider.
- 5. Predictive analysis — standardize features, split 80/20, tune Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV (cv=10), then compare accuracy and confusion matrices on the held-out test set.

Data Collection Methodology — SpaceX API

Key phrases and the flow of the API request

- Flow: GET <https://api.spacexdata.com/v4/launches/past> → `response.json()` → `pd.json_normalize()` → helper calls → filter to Falcon 9 → impute nulls → `dataset_part_1.csv`
- `requests.get(spacex_url)` — retrieve the past-launch records as JSON.
- `pd.json_normalize(response.json())` — flatten the nested JSON into a dataframe.
- `getBoosterVersion`, `getLaunchSite`, `getPayloadData`, `getCoreData` — call the rocket, launchpad, payload and core endpoints by ID to replace the IDs with real values.
- Filter: `data_falcon9 = data[data['BoosterVersion'] != 'Falcon 1']` — 90 launches remain.
- Missing values: PayloadMass nulls replaced with the column mean; LandingPad keeps its 26 nulls because 'no landing pad' is meaningful.

Data Collection Methodology — Web Scraping

Scraping the Falcon 9 launch table from Wikipedia

- Flow: `requests.get(static_url)` → `BeautifulSoup(html, 'html.parser')` → `find_all('table')` → parse header cells → row-by-row extraction → dataframe → `spacex_web_scraped.csv`
- `static_url` — the 9 June 2021 snapshot of 'List of Falcon 9 and Falcon Heavy launches'.
- `soup.title` returns: List of Falcon 9 and Falcon Heavy launches - Wikipedia — confirming the correct page was fetched.
- `html_tables = soup.find_all('table')`; the third table holds the launch records.
- `extract_column_from_header(th)` builds the column names, giving Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date and Time.
- Each row is walked with `find_all('td')`, the helper functions `date_time()`, `booster_version()`, `landing_status()` and `get_mass()` clean each cell, and the dictionary `launch_dict` is converted with `pd.DataFrame(launch_dict)`.

Data Wrangling Methodology

Cleaning, exploring and labeling the landing outcome

- Missing values: PayloadMass nulls replaced with the mean of the column; LandingPad nulls retained (26) because they encode 'no landing pad used'.
- Exploratory counts: launches per site (CCAFS SLC 40 = 55, KSC LC 39A = 22, VAFB SLC 4E = 13), occurrences of each orbit, and mission outcome per orbit type.
- Landing outcome label — the Outcome column holds eight values combining the result with the landing place (True ASDS, True RTLS, True Ocean, False ASDS, False RTLS, False Ocean, None ASDS, None None).
- bad_outcomes = {False ASDS, False Ocean, False RTLS, None ASDS, None None}; every outcome in that set becomes Class = 0 and everything else becomes Class = 1.
- Result: a clean binary target. `df['Class'].mean()` = 0.667, i.e. a two-thirds overall success rate across the 90 launches.
- Categorical features (Orbit, LaunchSite, LandingPad, Serial) were one-hot encoded with `pd.get_dummies`, producing an 80-column feature matrix cast to float64.

EDA and Data Visualization — Methodology

Charts plotted and why each one was chosen

- Scatter — Flight Number vs Payload Mass: shows that success improves with flight number and that very heavy payloads are rare.
- Scatter — Flight Number vs Launch Site: reveals which site handled the early (mostly failing) flights and which handled the later, more successful ones.
- Scatter — Payload Mass vs Launch Site: shows that VAFB SLC 4E has no launches above 10 000 kg.
- Bar — success rate by orbit type: identifies the orbits with perfect and with poor landing records.
- Scatter — Flight Number vs Orbit and Payload vs Orbit: separates a genuine orbit effect from an effect that is really about experience or payload weight.
- Line — yearly average success rate: the trend over time, which is the single clearest signal in the data.
- Scatter plots were chosen for pairwise relationships with a class color, bar charts for comparing a rate across categories, and a line chart for the time trend.

EDA with SQL

Summary of the SQL queries performed

- Displayed the names of the unique launch sites in the space mission.
- Displayed 5 records where the launch site name begins with 'CCA'.
- Displayed the total payload mass carried by boosters launched by NASA (CRS).
- Displayed the average payload mass carried by booster version F9 v1.1.
- Listed the date of the first successful landing outcome on a ground pad.
- Listed the boosters that had success on a drone ship and carried a payload between 4000 kg and 6000 kg.
- Listed the total number of successful and failed mission outcomes.
- Listed the booster versions that carried the maximum payload mass, using a subquery.
- Listed the failed drone-ship landings in 2015 with booster version, launch site and the month name.
- Ranked the count of landing outcomes between 2010-06-04 and 2017-03-20 in descending order.

Interactive Visual Analytics

Folium map and Plotly Dash — objects and why

- Folium — folium.Circle and folium.Marker with a DivIcon label mark every launch site, so the geographic spread of the sites is immediately visible.
- Folium — MarkerCluster groups the 56 launch records at their sites; each marker is colored green for success and red for failure, so a site's record can be read at a glance without over-plotting.
- Folium — folium.PolyLine plus a haversine distance function measures the distance from CCAFS SLC-40 to the coastline, railway, highway and nearest city, testing whether sites keep to safe proximities.
- Dash — dcc.Dropdown (site-dropdown) selects All Sites or one specific site.
- Dash — a pie chart callback shows either the share of successful launches per site or the success-versus-failure split for the selected site.
- Dash — dcc.RangeSlider (payload-slider, 0-10 000 kg) plus a scatter callback shows payload against landing outcome, colored by booster version category.

Predictive Analysis (Classification) — Methodology

- Target: `Y = df['Class']` (1 = first stage landed, 0 = did not land).
- Features: the 80-column one-hot encoded matrix, standardized with `StandardScaler().fit_transform(X)`.
- Split: `train_test_split(X, Y, test_size=0.2, random_state=2)` → 72 training rows and 18 test rows.
- Four algorithms tuned with `GridSearchCV(cv=10)`:
 - Logistic Regression — `C`, `penalty`, `solver`
 - Support Vector Machine — `kernel` (linear, rbf, poly, sigmoid), `C`, `gamma`
 - Decision Tree — `criterion`, `splitter`, `max_depth`, `max_features`, `min_samples_leaf`, `min_samples_split`
 - K-Nearest Neighbours — `n_neighbors`, `algorithm`, `p`
- Evaluation: `best_score_` for the validation accuracy, `.score(X_test, Y_test)` for the test accuracy, and a confusion matrix for the error profile of the best model.

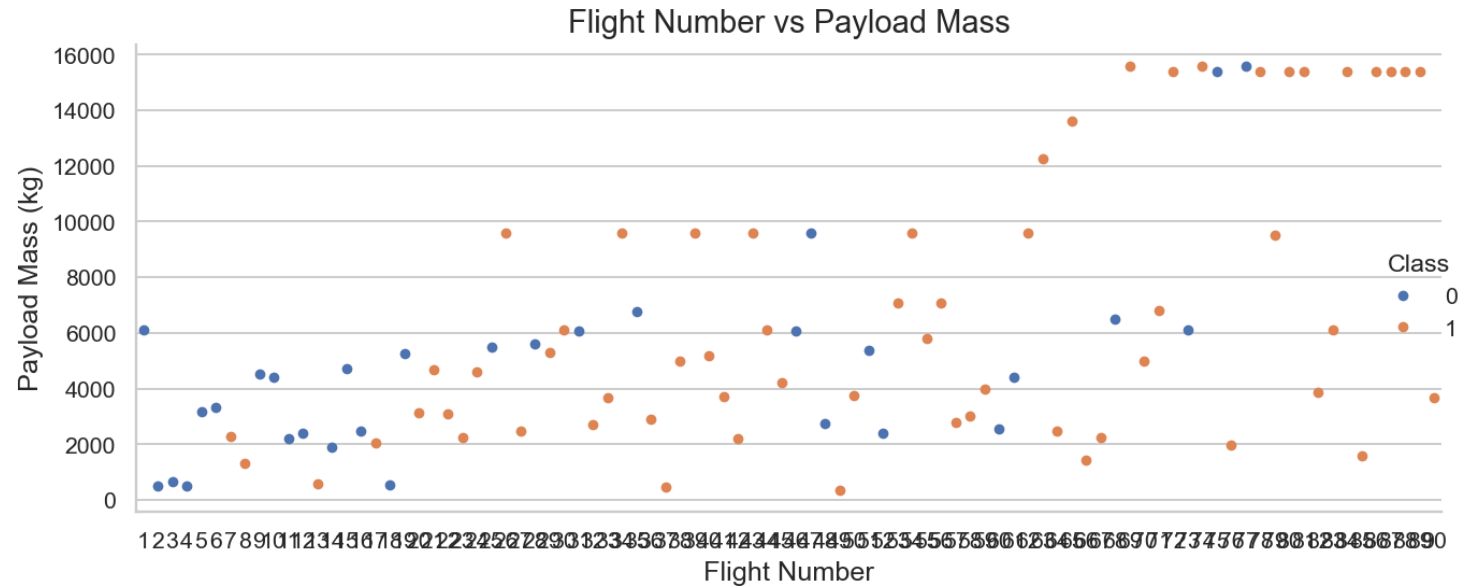
Results

EDA, SQL, interactive analytics and predictive modeling

- EDA with visualization results — 7 charts covering flight number, payload, launch site, orbit and the yearly trend
- EDA with SQL results — all 10 queries with the query text and the returned rows
- Folium map results — launch sites, clustered outcomes, expanded markers and proximity distances
- Plotly Dash dashboard results — the All Sites view and a single-site view
- Predictive analysis (classification) results — accuracy comparison, hyperparameters and the confusion matrix of the best model

Flight Number vs Payload Mass

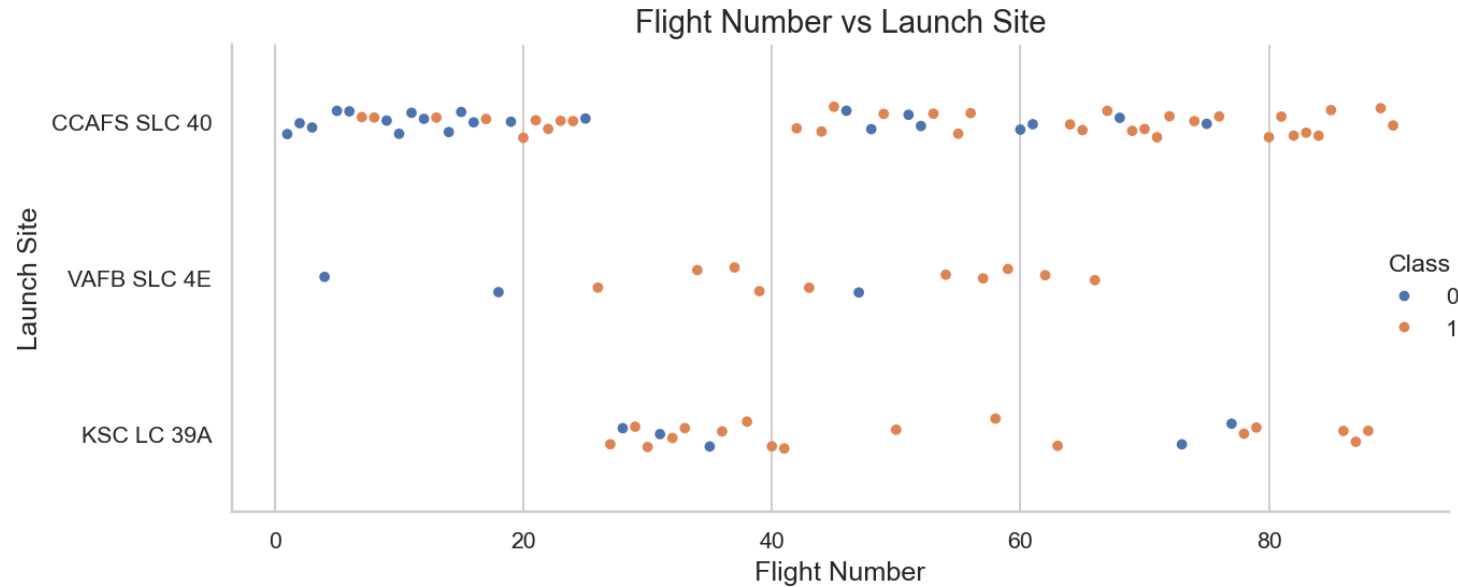
EDA with Visualization — Results



- What the chart shows: every launch plotted by flight number against payload mass, colored by outcome (orange = the first stage landed, blue = it did not).
- Finding: successful landings cluster on the right. Above flight 60 almost every launch lands, whatever the payload — so accumulated flight experience matters more than the weight being carried.
- Finding: payloads above 10 000 kg are rare, and the few that exist still land successfully, so heavy payload alone does not doom a landing.
- Implication for the model: FlightNumber should be a strong predictive feature.

Flight Number vs Launch Site

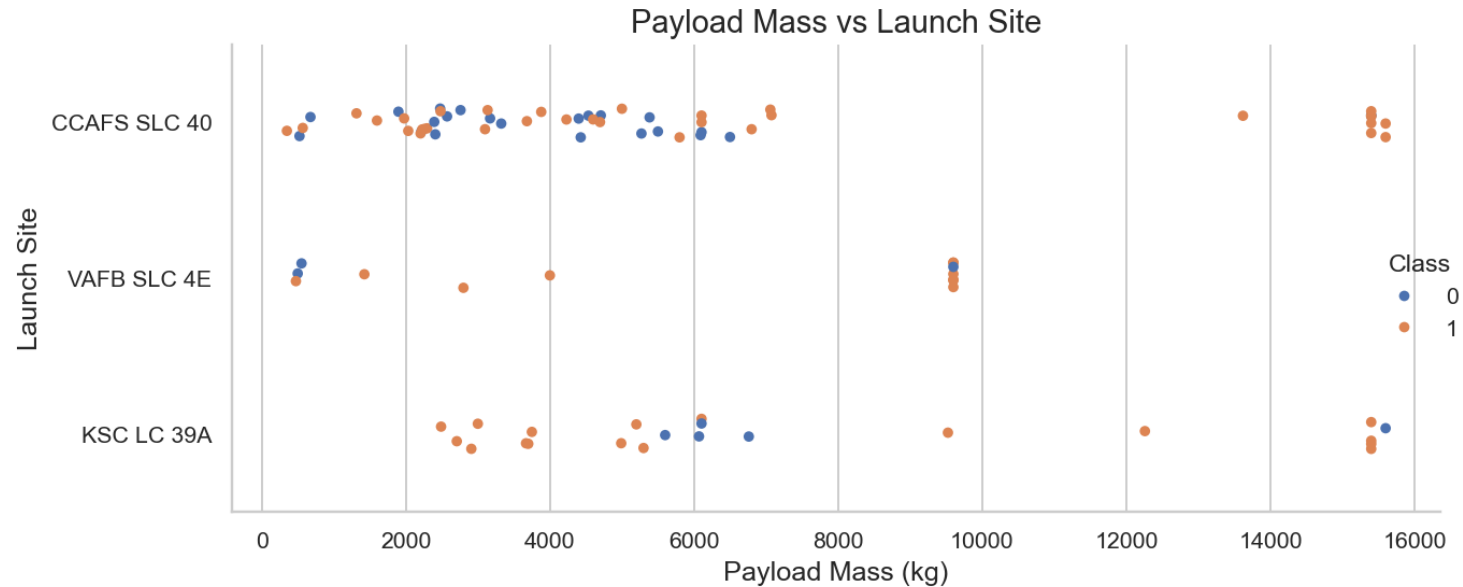
EDA with Visualization — Results



- What the chart shows: flight number against launch site, colored by landing outcome.
- Finding: CCAFS SLC 40 flew almost all of the first 25 missions, which is where the failures are concentrated. Its poor record is a consequence of history, not of the pad.
- Finding: KSC LC 39A only appears after flight 25 and is almost entirely successful, because it entered service once the landing technique had matured.
- Finding: VAFB SLC 4E flies sparsely throughout, with a mixed record.

Payload Mass vs Launch Site

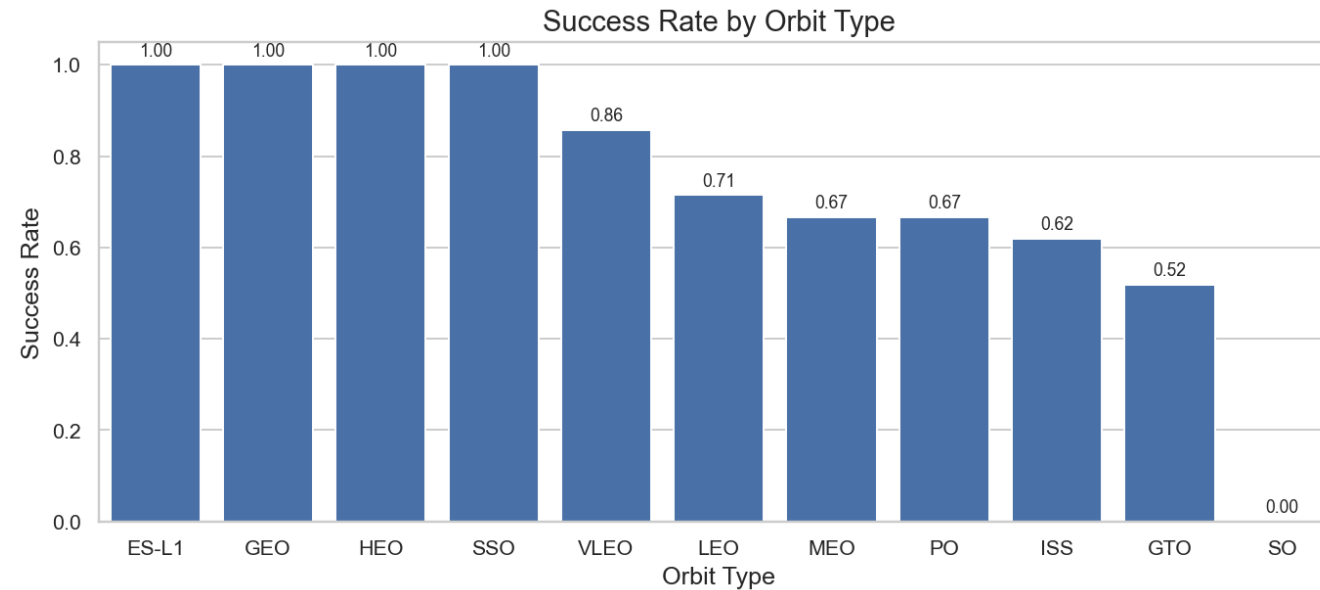
EDA with Visualization — Results



- What the chart shows: payload mass against launch site, colored by landing outcome.
- Finding: VAFB SLC 4E has no launch above 10 000 kg — the heaviest missions all fly from Florida.
- Finding: at the Florida pads, launches above roughly 7 000 kg land successfully more often than the mid-weight range, which is the opposite of the naive expectation.
- Implication: launch site and payload mass interact, so both belong in the feature set.

Success Rate by Orbit Type

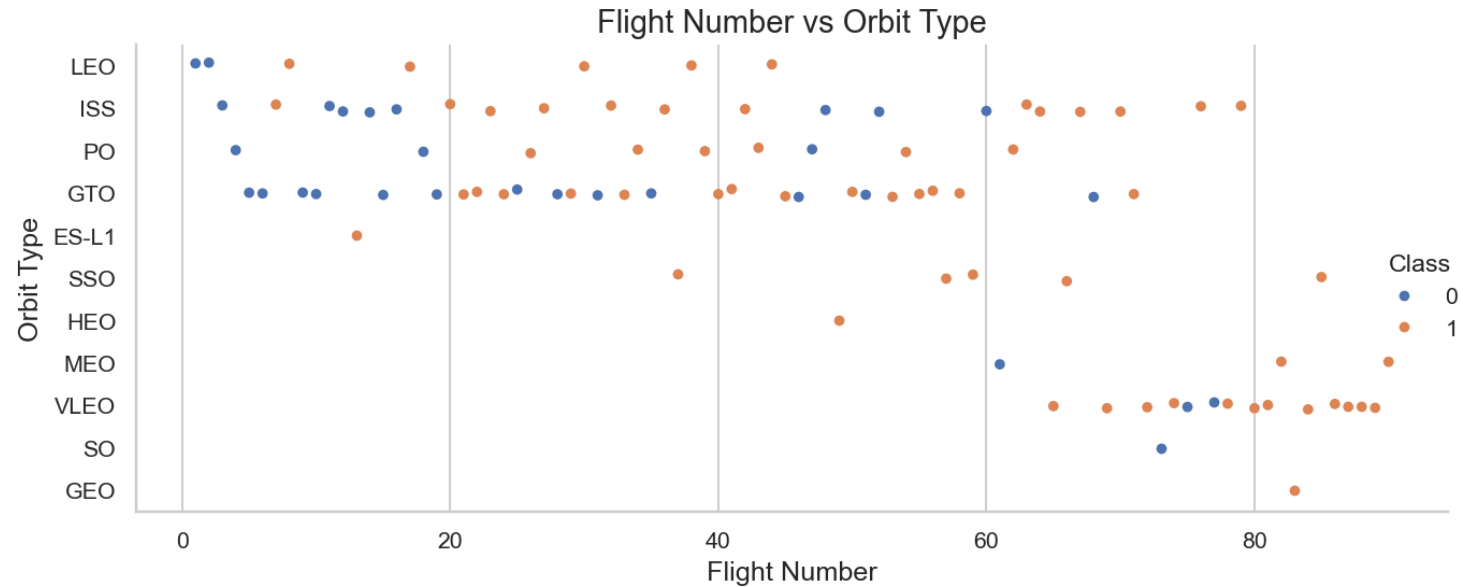
EDA with Visualization — Results



- What the chart shows: mean landing success rate for each orbit type.
- Finding: ES-L1, GEO, HEO and SSO record a 100% success rate; SO records 0%.
- Caveat: those perfect orbits have only a handful of launches each, so the rate is not statistically meaningful on its own.
- Finding: GTO, the busiest orbit, sits near 0.5 — high-energy transfer missions leave the least fuel margin for a landing burn.

Flight Number vs Orbit Type

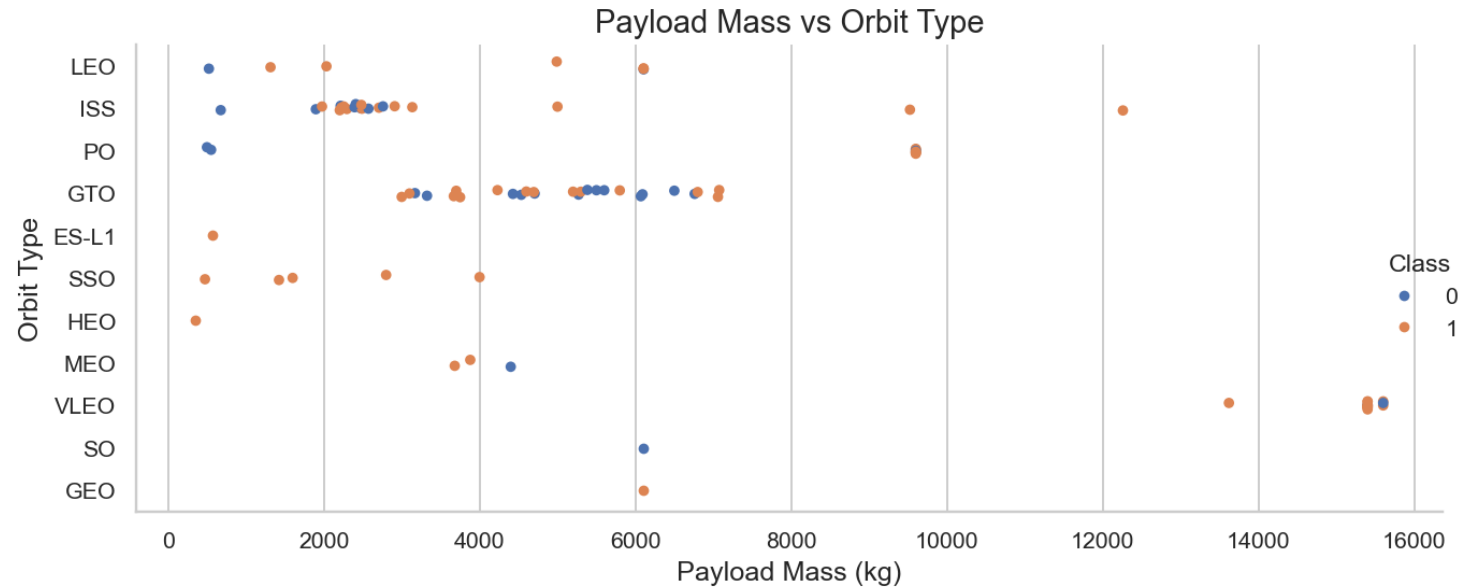
EDA with Visualization — Results



- What the chart shows: flight number against orbit type, colored by landing outcome.
- Finding: in LEO, success is clearly related to flight number — the later the mission, the more likely the landing.
- Finding: in GTO there is no such relationship; outcomes stay mixed across the whole flight range, confirming GTO is intrinsically harder.
- Finding: the 100% orbits appear only as isolated points, reinforcing the sample-size caveat from the previous slide.

Payload Mass vs Orbit Type

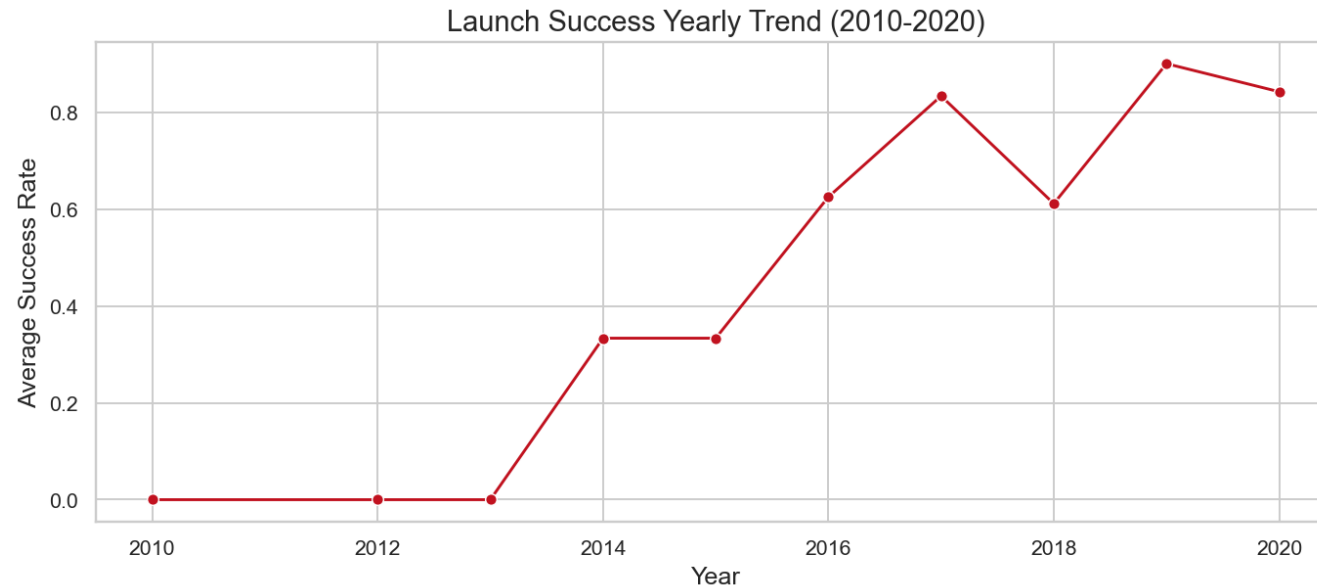
EDA with Visualization — Results



- What the chart shows: payload mass against orbit type, colored by landing outcome.
- Finding: heavy payloads land successfully in LEO, ISS and PO.
- Finding: for GTO the picture is mixed across every payload band, so payload mass alone does not explain the GTO outcomes.
- Implication: an orbit-by-payload interaction, not payload on its own, carries the predictive signal.

Launch Success Yearly Trend

EDA with Visualization — Results



- What the chart shows: the mean landing success rate for each year from 2010 to 2020.
- Finding: the rate is 0% until 2013, then climbs almost monotonically to roughly 80% by 2019-2020.
- Finding: the dip in 2018 interrupts the trend but does not reverse it.
- This is the single strongest signal in the dataset and it is temporal — SpaceX solved the landing problem by iteration.

All Launch Sites

EDA with SQL — Results and Queries

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Task 1 — Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Result: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E — four distinct pads.
- Three are in Florida and one in California, which is why the Folium map later shows two geographic clusters.
- DISTINCT is needed because the table stores one row per launch, so each pad appears many times.
- These four values become the launch-site categories that are one-hot encoded for the classification models.

Launch Site Names Beginning with 'CCA'

EDA with SQL — Results and Queries

```
SELECT Date, Launch_Site, Booster_Version, Payload, PAYLOAD_MASS__KG_  
FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Task 2 — 5 records where launch site begins with 'CCA'

Date	Launch_Site	Booster_Version	Payload	PAYLOAD_MASS__KG_
2010-06-04	CCAFS LC-40	F9 v1.0 B0003	Dragon Spacecraft Qualification	0
2010-12-08	CCAFS LC-40	F9 v1.0 B0004	Dragon Demo flight C1, two CubeSats, barrel	0
2012-05-22	CCAFS LC-40	F9 v1.0 B0005	Dragon demo flight C2	525
2012-10-08	CCAFS LC-40	F9 v1.0 B0006	SpaceX CRS-1	500
2013-03-01	CCAFS LC-40	F9 v1.0 B0007	SpaceX CRS-2	677

- The LIKE wildcard matches both Cape Canaveral pads.
- Result: the five earliest Cape Canaveral missions, all F9 v1.0 boosters with small payloads — the Dragon demonstration flights that opened the program.

Total Payload Mass — NASA (CRS)

EDA with SQL — Results and Queries

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass  
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Task 3 — Total payload mass carried by boosters from NASA (CRS)

Total_Payload_Mass_KG
45596

- Result: 45 596 kg carried in total for NASA's Commercial Resupply Services.
- CRS is the single largest customer block in the dataset, which is why ISS-bound missions dominate the LEO/ISS orbit categories.

Average Payload Mass — F9 v1.1

EDA with SQL — Results and Queries

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass  
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

Task 4 — Average payload mass carried by booster version F9 v1.1

Avg_Payload_Mass_KG
2928.4

- Result: 2 928.4 kg average payload for the F9 v1.1 booster.
- That is well below the fleet average — v1.1 was the early, less capable variant, and it also has the worst landing record.

First Successful Ground-Pad Landing

EDA with SQL — Results and Queries

```
SELECT MIN(Date) AS First_Successful_Ground_Landing  
FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

Task 5 — Date of the first successful landing outcome on ground pad

First_Successful_Ground_Landing
2015-12-22

- Result: 2015-12-22 — the first time a first stage returned to a ground pad intact.
- This date is the inflection point of the yearly success trend seen in the EDA charts.

Successful Drone-Ship Landings, 4000-6000 kg

EDA with SQL — Results and Queries

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Task 6 — Boosters: success on drone ship, payload 4000-6000 kg

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

- Result: F9 FT B1022, B1026, B1021.2 and B1031.2.
- All are Full Thrust boosters, and two are re-flown cores (the .2 suffix) — evidence that re-used boosters land as reliably as new ones.

Total Successful and Failed Mission Outcomes

EDA with SQL — Results and Queries

```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Task 7 — Total number of successful and failed mission outcomes

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Result: 98 'Success', plus 1 'Success' with a trailing space and 1 'Success (payload status unclear)', against a single 'Failure (in flight)'.
- The duplicated label is a data-quality artifact worth noting — a strict GROUP BY splits it out instead of merging it.
- Mission success and landing success are very different problems: the rocket almost always delivers the payload, even when the booster is lost.

Boosters Carrying the Maximum Payload

EDA with SQL — Results and Queries

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Task 8 — Booster versions that carried the maximum payload mass

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

- A subquery finds the maximum payload, then the outer query names the boosters.
- Result: twelve boosters, every one of them a B5 core — only the final Block 5 design lifts the heaviest payloads.

2015 Failed Drone-Ship Landings

EDA with SQL — Results and Queries

```
SELECT CASE substr(Date, 6, 2) WHEN '01' THEN 'January' ... END AS Month,  
       Landing_Outcome, Booster_Version, Launch_Site  
FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015'  
       AND Landing_Outcome = 'Failure (drone ship)';
```

Task 9 — 2015 failure landing outcomes on drone ship (month, booster, site)

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- substr() extracts the month and a CASE statement converts it to a name.
- Result: two failures, January and April 2015, both F9 v1.1 boosters from CCAFS LC-40 — the drone ship attempts that preceded the first successful landing.

Landing Outcomes Ranked, 2010-06-04 to 2017-03-20

EDA with SQL — Results and Queries

```
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

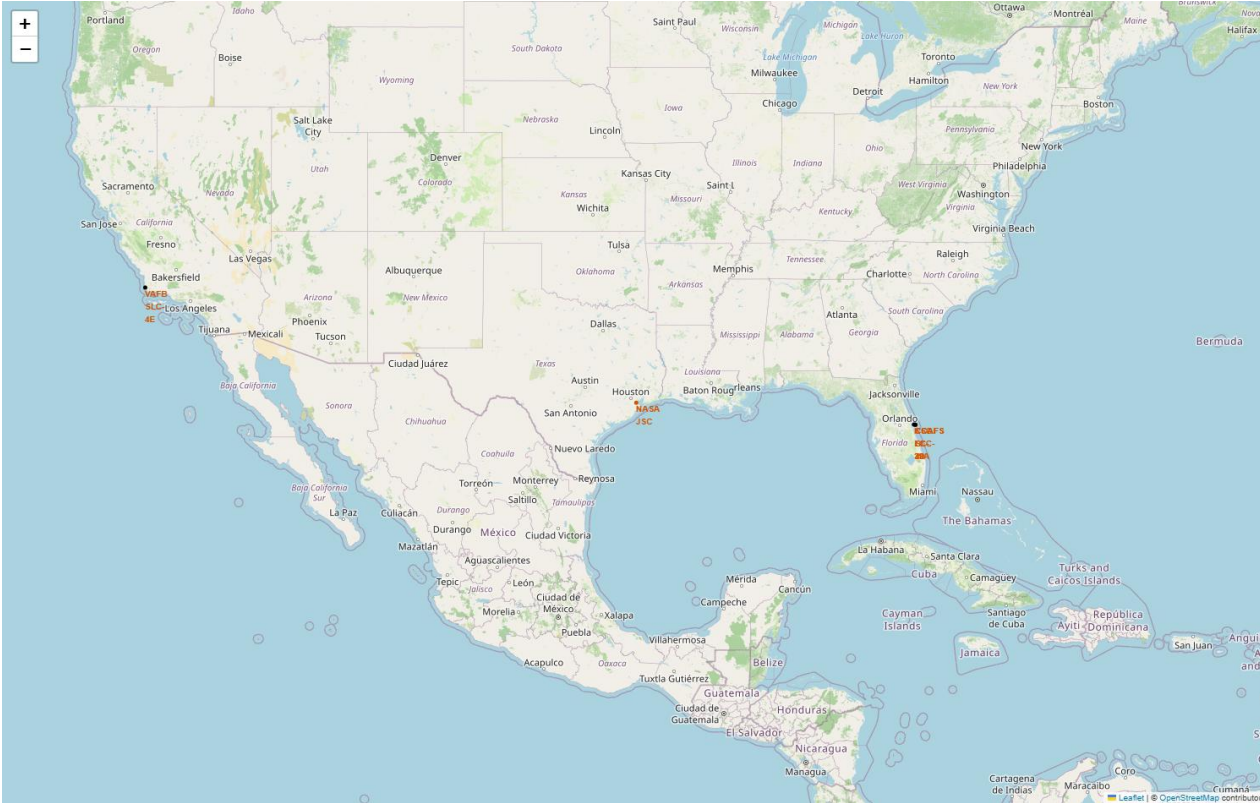
Task 10 — Landing outcomes ranked, 2010-06-04 to 2017-03-20

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Result: 'No attempt' leads with 10, followed by drone ship success and failure at 5 each, then ground pad success at 3.
- In the program's first seven years the most common outcome was not even trying to land — the recovery capability simply did not exist yet.

Launch Sites on a Folium Map

Folium Map — Results



- Objects used: folium.Circle plus folium.Marker with a Divicon text label at each of the four launch sites, on a map centered on NASA Johnson Space Center.
- Finding: all four sites sit on a coast — three on the Atlantic in Florida, one on the Pacific in California. A failed launch therefore falls over water, not over land.
- Finding: the Florida sites are at roughly 28.5° N, closer to the equator than Vandenberg at 34.6° N, so they gain more free velocity from the Earth's rotation.
- Finding: no launch site is inland, which constrains where a competitor could build.

Launch Records Clustered by Site

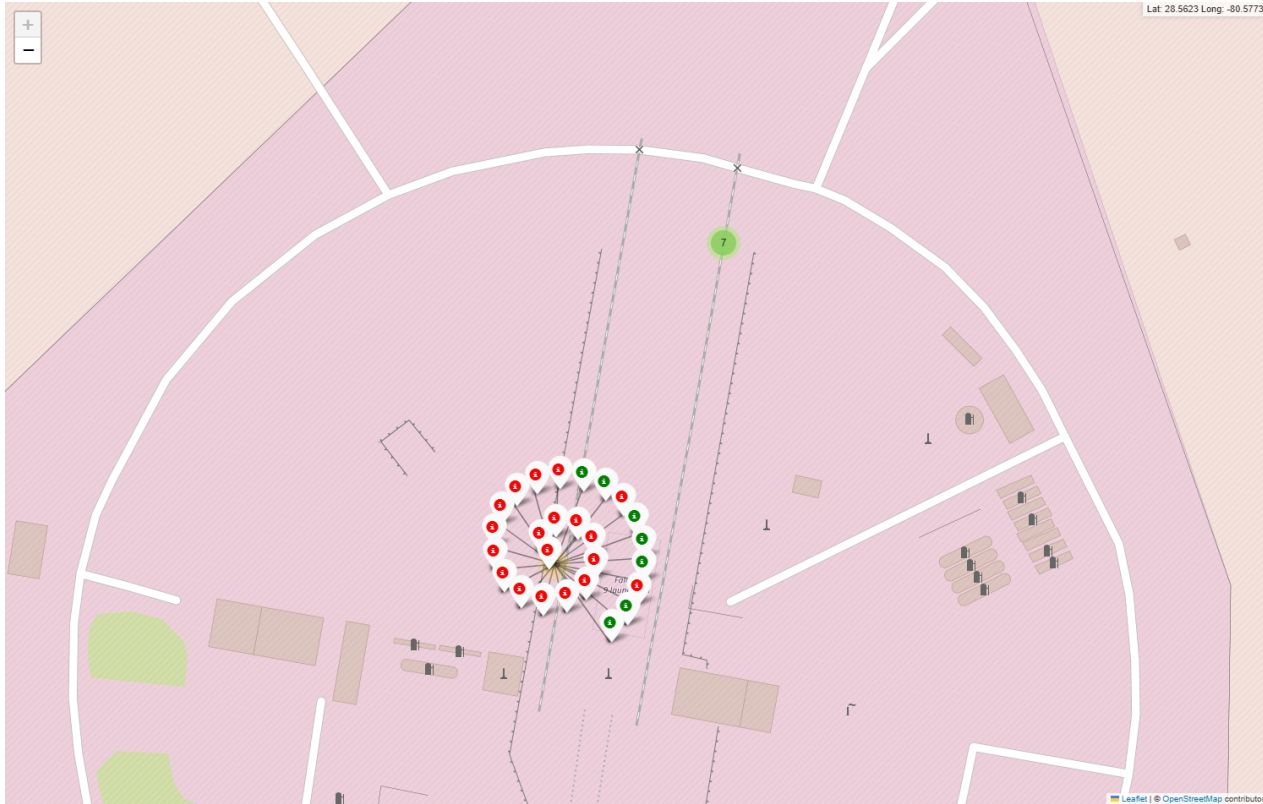
Folium Map — Results



- Objects used: a MarkerCluster holding all 56 launch records, each colored green for a successful landing and red for a failure.
- Finding: the cluster counts make the launch volume obvious at a glance — the Florida cluster dwarfs Vandenberg.
- Clustering is what makes this readable: the records at one pad share identical coordinates, so without it every marker would sit on top of the others.

Success and Failure Markers at CCAFS SLC-40

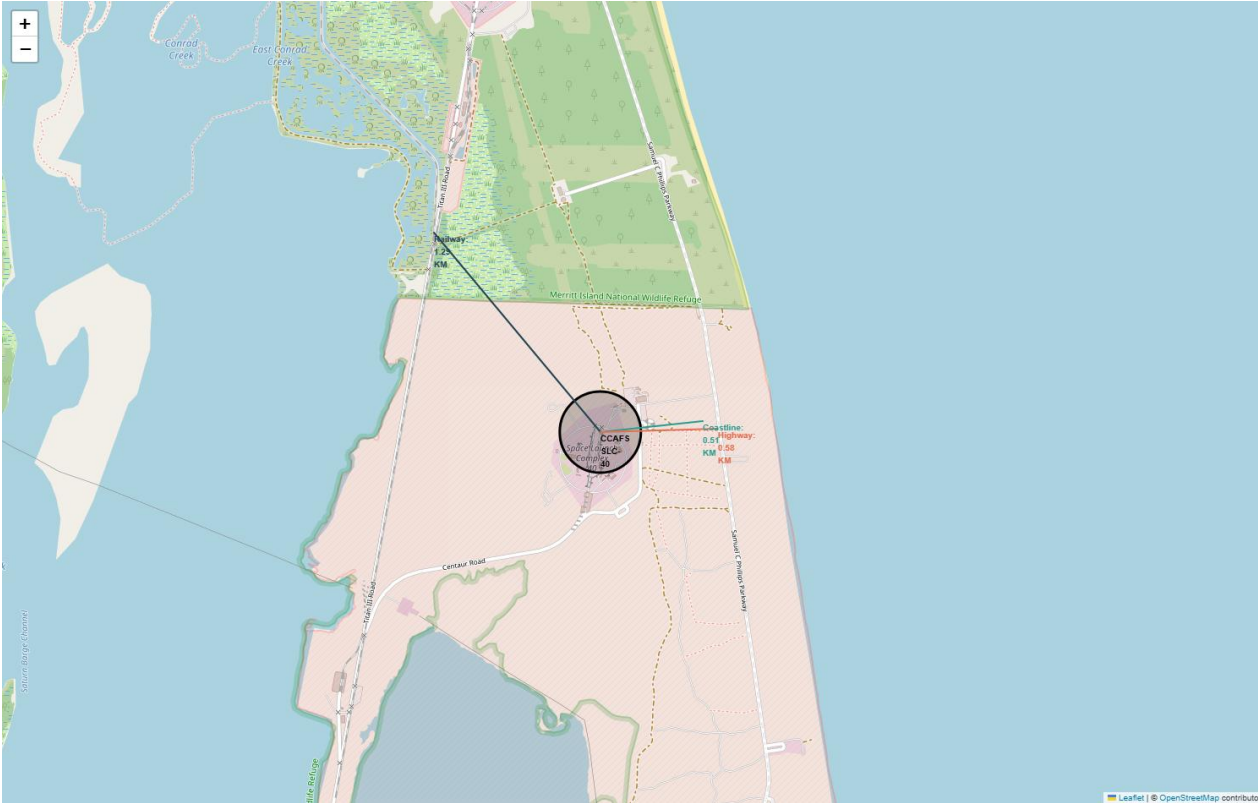
Folium Map — Results



- The cluster is expanded to spiderfy the individual markers at CCAFS SLC-40.
- Finding: red markers clearly outnumber green at this pad, matching the EDA result that CCAFS SLC-40 flew the early, failure-prone missions.
- Finding: the green markers are the later flights, so the pad's record improves over time rather than being uniformly poor.
- This visual confirms that launch site alone is a misleading predictor unless flight number is considered alongside it.

Launch Site Proximities — CCAFS SLC-40

Folium Map — Results

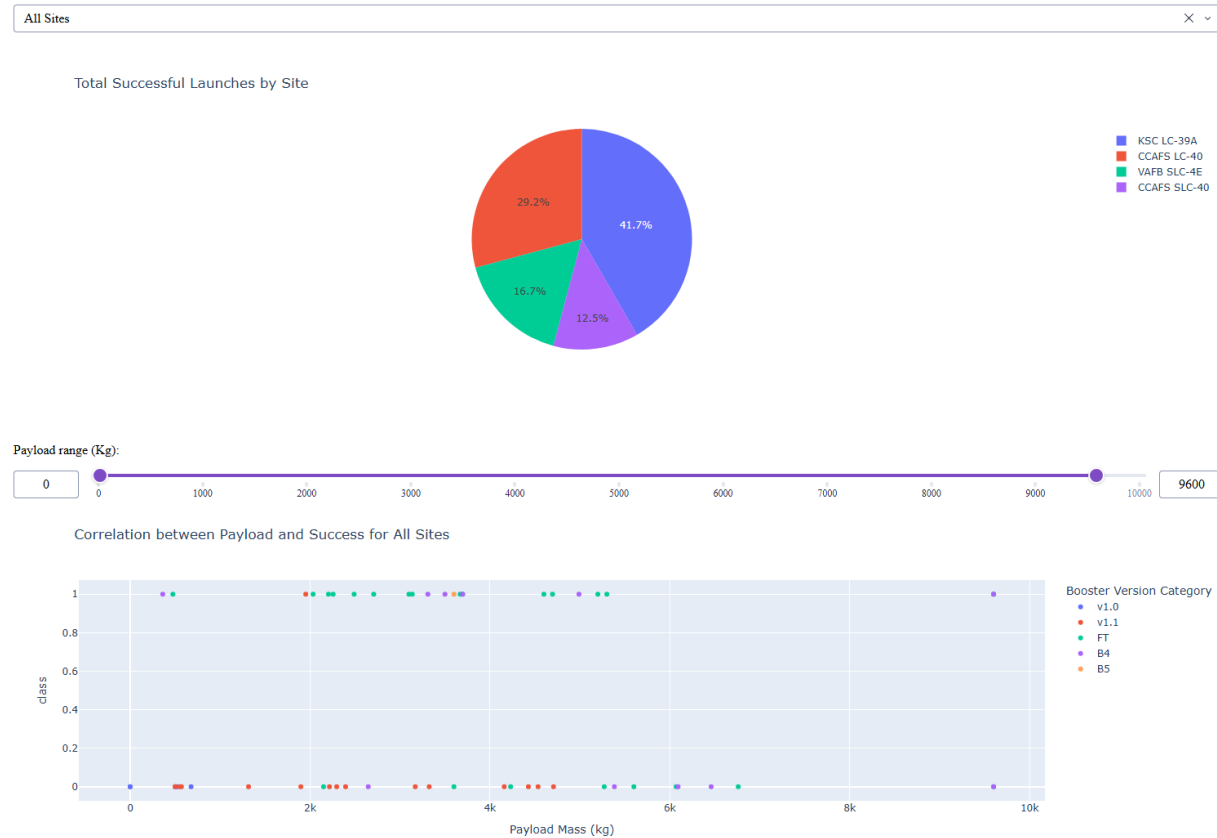


- Objects used: folium.PolyLine from CCAFS SLC-40 to each point of interest, with the haversine distance calculated in Python and drawn as a marker label.
- Results: coastline 0.51 km, highway 0.58 km, railway 1.29 km, nearest city (Titusville) 23.19 km.
- Finding: launch sites keep very close to transport links, which are needed to move stages and propellant to the pad.
- Finding: they stay far from population centers — over 23 km in this case — which is the safety constraint that dictates site selection.

Plotly Dash Dashboard — All Sites

Plotly Dash Dashboard — Results

SpaceX Launch Records Dashboard



- Dashboard components on screen: the dcc.Dropdown set to 'All Sites', the success pie chart, the dcc.RangeSlider covering 0-10 000 kg, and the payload-versus-outcome scatter chart.

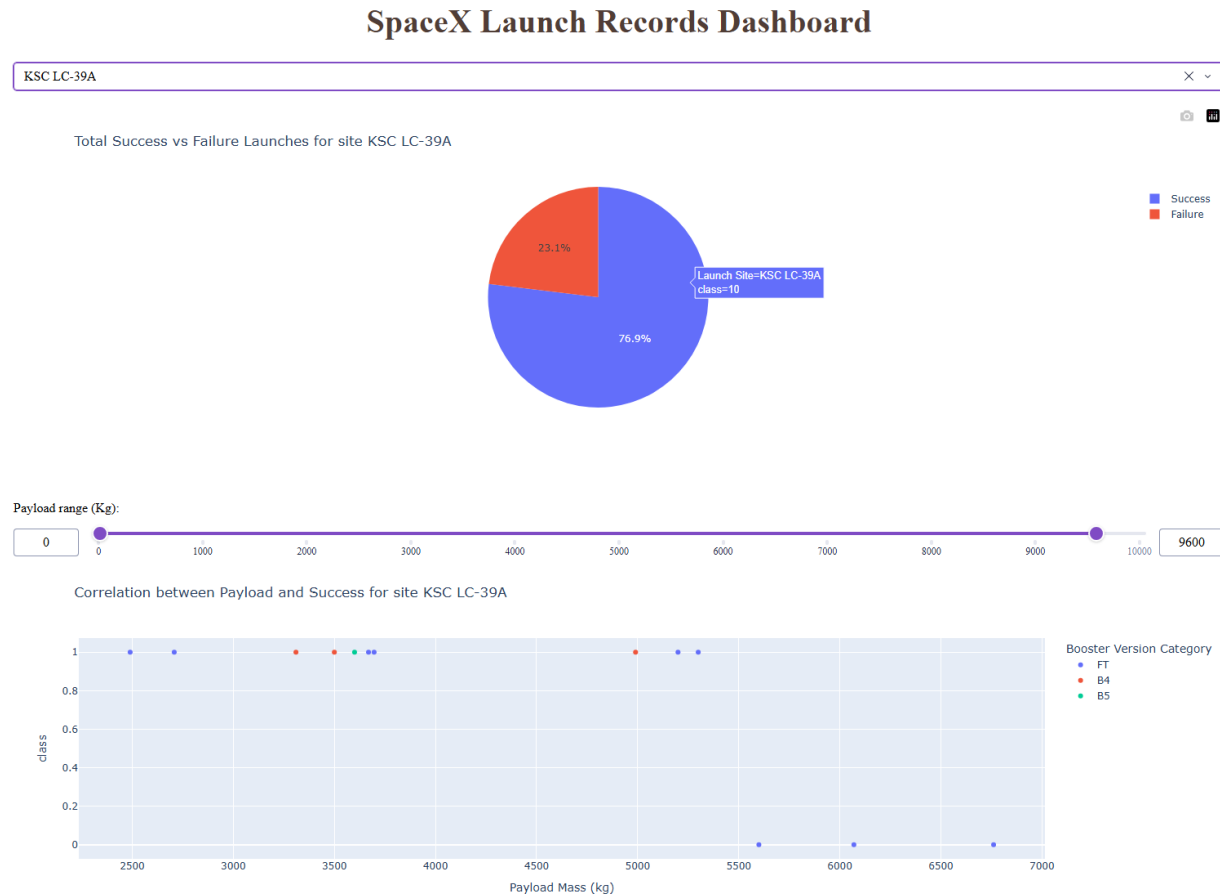
- Pie result: KSC LC-39A contributes 41.7% of all successful launches, CCAFS LC-40 29.2%, VAFB SLC-4E 16.7% and CCAFS SLC-40 12.5%.

- Scatter finding: successes concentrate in the 2 000-5 500 kg band, which lands 55% of the time (18 of 33). The 6 000-10 000 kg band lands only 33% (3 of 9) and payloads under 2 000 kg only 27% (3 of 11).

- Scatter finding: by booster version, FT and B4 cores account for 22 of the 24 successes; v1.0 never lands (0 of 5) and v1.1 lands once in 15 attempts.

Plotly Dash Dashboard — KSC LC-39A Selected

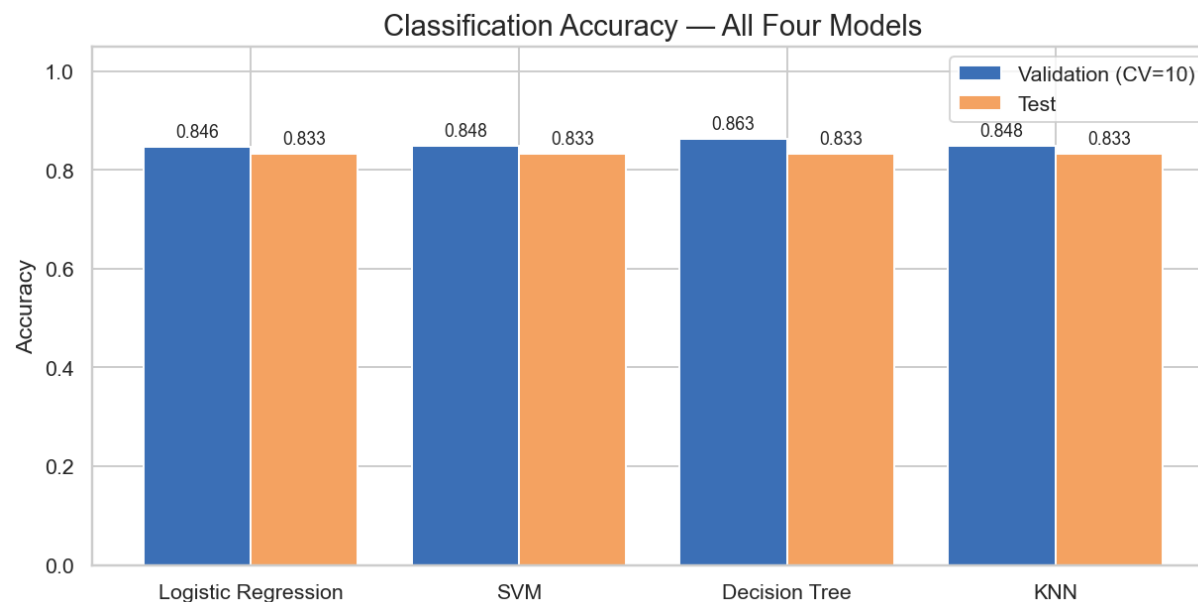
Plotly Dash Dashboard — Results



- Selecting KSC LC-39A in the dropdown fires both callbacks: the pie switches to that site's success-versus-failure split and the scatter filters to its launches only.
- Pie result: KSC LC-39A lands roughly three quarters of its first stages — the best ratio of any site in the dataset.
- Scatter finding: its successes concentrate in the lighter payload range and are flown almost entirely by FT and B4 boosters.
- Interaction check: the same dropdown drives two independent Output components, which is the callback behavior the lab asks for.

Classification Accuracy — All Models

Predictive Analysis (Classification) — Results



- Validation accuracy (10-fold cross-validation on the 72 training rows): Decision Tree 0.8625, SVM 0.8482, KNN 0.8482, Logistic Regression 0.8464.
- Test accuracy on the held-out 18 launches: all four models score 0.8333, i.e. 15 correct out of 18.
- The tie is a property of the test set, not of the models — with 18 rows the accuracy can only move in steps of 5.6 percentage points, so different models easily land on the same score.
- The cross-validated score is therefore the fairer ranking, and on that measure the Decision Tree is ahead by about 1.4 points.

Model Comparison Table

Predictive Analysis (Classification) — Results

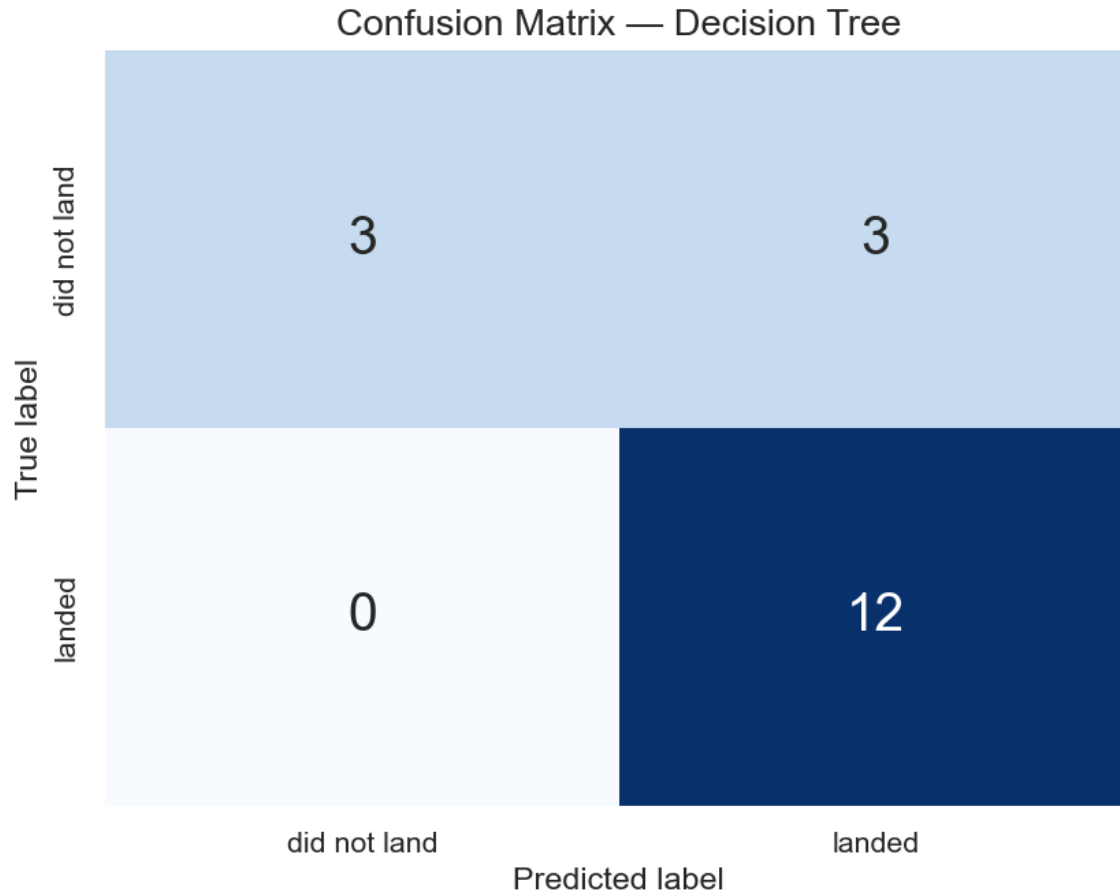
Model comparison — validation vs test accuracy

Model	Best validation accuracy (CV=10)	Test accuracy
Logistic Regression	0.8464	0.8333
SVM	0.8482	0.8333
Decision Tree	0.8625	0.8333
KNN	0.8482	0.8333

- Logistic Regression: validation 0.8464, test 0.8333 — best hyperparameters {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
- SVM: validation 0.8482, test 0.8333 — best hyperparameters {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}
- Decision Tree: validation 0.8625, test 0.8333 — best hyperparameters {'criterion': 'gini', 'max_depth': 6, 'max_features': 'sqrt', 'min_samples_split': 2}
- KNN: validation 0.8482, test 0.8333 — best hyperparameters {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}
- Every model was tuned by GridSearchCV over the parameter grid listed in the methodology section, refitted on the full training set, then scored once on the untouched test set.

Confusion Matrix — Decision Tree

Predictive Analysis (Classification) — Results



- Counts on the 18 test launches: 12 true positives, 3 true negatives, 3 false positives, 0 false negatives.
- Recall on the landed class is 1.00 — the model never misses a successful landing. Precision on that class is $12/15 = 0.80$.
- Every error is the same kind: a booster the model expects to land that in fact does not. The model has no way to be pessimistic.
- Business consequence: a bid priced from this model would systematically under-estimate cost. Weighting the loss function against false positives, or raising the decision threshold, is the obvious next iteration.

Best Performing Model

- Best model: Decision Tree — validation accuracy 0.8625, test accuracy 0.8333.
- All four models score identically on the test set because the test set holds only 18 launches — a single reclassified launch moves accuracy by 5.6 percentage points. The validation score across 10 folds is the more reliable comparison.
- The Decision Tree is also the most interpretable of the four, which matters for a cost model that has to be explained to a bidding team.
- Caveat: with 90 rows and 80 features the models are working in a very wide, very shallow space. More launches, or aggressive feature selection, would firm up these results.

Conclusion

- Launch success improves with experience: the yearly success rate climbs from 0% in 2010-2013 to about 80% by 2019-2020.
- Launch site matters. KSC LC-39A has the best record; CCAFS SLC 40 carries the early failures simply because it flew the early missions.
- Orbit matters. ES-L1, GEO, HEO and SSO show 100% success in this dataset, while GTO — the busiest orbit — sits near 50%.
- Payload matters, but less than expected: lighter payloads (< 4000 kg) land successfully more often, yet the very heaviest payloads in LEO, ISS and PO still land.
- Prediction: with only 90 launches, all four tuned classifiers reach 83.33% test accuracy. The Decision Tree has the best cross-validated score and is the model to carry forward.
- Business meaning: a bidder can estimate launch cost from site, orbit, payload and booster version before the launch is flown — which is exactly what Space Y needs to price a competing bid.

Innovative Insights

What the data shows beyond the required analysis

- The strongest predictor is not physical but temporal. Flight number — a proxy for accumulated engineering experience — separates the classes better than payload mass does. SpaceX's landing problem was solved by iteration, not by flying easier missions.
- Orbit success rates are partly an artefact of volume. The orbits at 100% (ES-L1, GEO, HEO, SSO) each have only a handful of launches; GTO looks worse largely because it has the most attempts. Any orbit-based rule needs a sample-size caveat.
- The confusion matrix reveals an asymmetric risk. The model never misses a success but over-predicts three failures as successes, so a bid priced from this model would be too optimistic. Weighting the loss toward false positives would be the next iteration.
- The launch sites cluster tightly against the coast, close to a highway and railway but over 20 km from the nearest town — a geographic pattern that a competitor choosing a new site would have to replicate.

Appendix

- Notebooks: data collection (API), data collection (web scraping), data wrangling, EDA with visualization, EDA with SQL, Folium interactive map, and machine-learning prediction.
- Dashboard: `spacex_dash_app.py` — Plotly Dash application with a launch-site dropdown, a success pie chart, a 0-10 000 kg payload range slider, and a payload-versus-outcome scatter chart.
- Data sources: SpaceX REST API v4 (api.spacexdata.com/v4/launches/past) and the Wikipedia page 'List of Falcon 9 and Falcon Heavy launches' (9 June 2021 snapshot).
- Libraries: pandas, numpy, matplotlib, seaborn, beautifulsoup4, requests, sqlite3 / ipython-sql, folium, plotly, dash and scikit-learn.
- Dataset size after wrangling: 90 Falcon 9 launches, 80 features after one-hot encoding, 72 training rows and 18 test rows.